

SAWSAN ABDULBARI

Data Scientist & Machine Learning Engineer | AI Innovation

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BIOGRAPHY

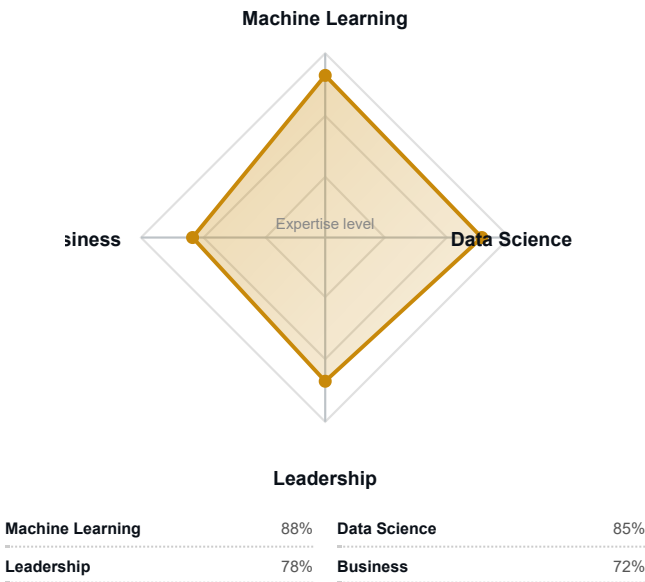
I build full-stack AI and ML systems that ship production machine learning, generative AI, and data software used by real teams to drive measurable business outcomes and automation. Before moving into engineering, I spent over a decade as an operator including 7 years as a founder, plus senior management and quality control roles. That experience shapes how I build AI today: focused on real-world impact, adoption, and audit-ready governance rather than accuracy in isolation. **Trilingual professional** (Arabic, English, Finnish) with **entrepreneurial background** and **passion for climate change solutions**.

Professional overview: I have contributed to environmental sustainability by transforming plastic waste into valuable resources. Throughout my career, I managed supply operations and spearheaded new business initiatives, and focused on developing eco-friendly solutions that align with climate change mitigation efforts.

Commitment to climate change: I am passionate about incorporating climate change awareness into every aspect of business operations and life. My ongoing commitment involves adopting and advocating for practices that minimize environmental footprints, enhance recycling processes, and drive progress toward a more sustainable planet. My approach ensures that business growth and environmental health go hand in hand.

Key Differentiators: Global collaboration (distributed teams in global contributor projects), Production Expertise (deployed ML models), Business Knowledge (7-year entrepreneurial experience), Climate Focus (environmental sustainability advocate), Academic Excellence (student mentor & curriculum developer).

EXPERTISE AND TOPICS



Relative scale aligned with portfolio skills visualization (self-assessment).

Topics: Artificial intelligence, machine learning, data science, generative AI, RAG systems, data analytics, business intelligence, climate change AI.

WORK EXPERIENCE

Data Scientist and Machine Learning Engineer

Anticipated Start: Dec, 2026

University of Helsinki | Helsinki, Finland

Anticipated role: data science and machine learning engineering at the University of Helsinki.

Data Science Machine Learning Python

Junior Expert specializing in AI

Aug 2026 - Present

Demos Helsinki | Helsinki Metropolitan Area · Hybrid

Applying AI expertise to Demos Helsinki: Connected Knowledge System (CKS), contributing to the development and delivery of AI-powered knowledge solutions.

Artificial Intelligence Machine Learning Connected Knowledge System

Data Scientist and Machine learning Engineer

Jun 2026 - Aug 2026

Dashtera | Finland

Built and deployed data science and machine learning solutions, applying end-to-end ML workflows to support Dashtera's products and services.

Data Science Machine Learning Python

Machine Learning Engineer

2024 - Present

Omdena | Palo Alto, CA (Remote)

Collaborating on AI innovation in globally distributed teams. Researching and implementing RAG architecture for enterprise applications.

Python RAG Architecture Machine Learning AI Research

Student Assistant

2024 - 2025

Häme University of Applied Sciences (HAMK) | Finland

Improved educational outcomes for 50+ students and developed curriculum. Provided personalized guidance and technical mentoring.

Python Teaching Curriculum Development

Lead Machine Learning Engineer

2024

Omdena Kitwe, Zambia | Remote

Delivered a COVID-19 forecasting system for the Kitwe, Zambia (Omdena) chapter: XGBoost models, Streamlit app, and standard regression metrics (MAE, RMSE, R²) as in the public repository. Worked with a global team of 15+ contributors.

Team Leadership XGBoost Healthcare AI

Machine Learning Engineer

2023 - 2024

Raqameyyah Digital Consultancy | Egypt (Remote)

Built and deployed ML models using advanced techniques. Developed chatbots with advanced prompt engineering and created front-end applications to enhance customer experience.

Machine Learning Feature Engineering Application Deployment Python

Data Analyst

2023

Raqameyyah Digital Consultancy | Egypt (Remote)

Cleaned and preprocessed large datasets for analysis. Developed applications for data filtering and visualization and created automated data processing pipelines.

Data Analysis Dashboards Streamlit EDA

Founder

2007 - 2014

Plastic Recycling for Sustainable Solutions | Hybrid

Founded and managed a plastic recycling factory focused on environmental protection. Developed business strategies and operational methods and managed employees and production processes. Promoted environmental protection by transforming plastic waste into valuable resources.

Business-to-Business (B2B) New Business Development Supply Operations Environmental Sustainability

EDUCATION

Bachelor of Business Information Technology (Data Science and Machine learning) — Grade 4.7

Häme University of Applied Sciences (HAMK), Finland

2023 - 2026

Specialization in Data Science and Business Intelligence. Focus on machine learning, data analytics, and business applications.

Thesis (BITCA): Microsoft-based performance goal management system for strategy implementation. PDF on the portfolio site. Catalogue: [HAMK Finna \(vanaicat.144542\)](#).

Climate Change AI (CCAI) Summer School

Climate Change AI (CCAI)

2024

Intensive program on machine learning and AI for climate change mitigation, adaptation, and environmental decision-making. Certificate:

Climate_Change_AI_Sawsan_Abdulbari.pdf on the portfolio site.

Google Data Analytics Specialization

Google

Nov 2023

Courses: Foundations: Data, Data, Everywhere; Ask Questions to Make Data-Driven Decisions; Prepare Data for Exploration; Process Data from Dirty to Clean; Analyze Data to Answer Questions; Share Data Through the Art of Visualization; Google Data Analytics Capstone: Complete a Case Study. Certificate PDF: *Google_Data_Analytics.pdf* on the portfolio site.

Vocational Qualification in Business (Business Administration)

Tampere Vocational College (Tredu), Finland

2020 - 2022

Business administration and management fundamentals. Project management and organizational skills.

LANGUAGES

Arabic

Native

English

Professional proficiency

Finnish

Professional proficiency

PROJECT REFERENCES

Demos Helsinki: Connected Knowledge System (CKS)

Artificial Intelligence, Machine Learning, AI Research

AI-powered knowledge system for Demos Helsinki research and innovation, supporting the development and delivery of AI-powered knowledge solutions.

Kiva Loan EDA Dashboard

Python, Pandas, Plotly, Streamlit

Interactive dashboard analyzing Kiva microloan data with comprehensive EDA features. Includes loan distribution analysis, country-wise insights, and borrower demographics.

California Housing Price Predictor

Python, Scikit-learn, Pandas, Streamlit

Machine learning model predicting California housing prices using regression algorithms. Features data preprocessing, feature engineering, and model comparison.

NYC Taxi Trip Duration Prediction

Python, XGBoost, Folium, Streamlit

Predicts taxi trip duration in NYC using machine learning algorithms. Incorporates geospatial features, weather data, and traffic patterns.

E-commerce Analytics Dashboard

Python, Pandas, Plotly, Streamlit

Comprehensive e-commerce analytics dashboard providing insights into sales performance, customer behavior, and product trends with interactive visualizations.

News Sentiment & Keyword Analysis Dashboard

Python, NLTK, Pandas, Streamlit

Comprehensive dashboard for analyzing news sentiment and extracting key insights from articles using NLTK VADER sentiment analysis and advanced NLP techniques.

Carbon Fraud Detection Network

Python, FastAPI, Google Cloud, Vertex AI, PostgreSQL, Redis

Advanced AI-powered system for detecting carbon fraud using personality-based consensus agents and multi-source data integration. Features real-time fraud detection, adversarial AI agents, and comprehensive emission validation through CLIMATIQ API, OpenWeather, and satellite data integration.

PIRHA Pre-Procedure Rehabilitation Guide

Vercel, Web, Healthcare UX

Web guide for pre-procedure rehabilitation for the Pirkanmaa wellbeing region (PIRHA). Supports patients and families before medical procedures. Live at [ict-project-pirha.vercel.app](https://project-pirha.vercel.app).

Trading Companion — discipline system for traders

Vite, React, TypeScript, Supabase, Tailwind, Stripe, i18next, Vitest, Playwright

Trading app: position sizing, risk and discipline tools, plans and strategies, trade history, Stripe, PWA, automated tests and monitoring. <https://data-diwan-trading.vercel.app/>

COVID-19 Case Prediction System

Python, XGBoost, scikit-learn, Streamlit, pandas

COVID-19 imputed cases and deaths for the Kitwe, Zambia (Omdena) chapter: trained XGBoost models, multi-page Streamlit app, and MAE/RMSE/R² evaluation as documented on GitHub.

Customer Segmentation Dashboard

Python, Scikit-learn, Plotly, Streamlit

Advanced customer segmentation using K-means clustering and RFM analysis. Interactive dashboard for business insights with actionable recommendations.

Bank Note Authentication

Python, scikit-learn, Streamlit, pandas

UCI banknote (four wavelet features): compared SVM, k-NN, logistic regression, and decision trees, with cross-validation and a Streamlit deployment per the public repository.

Smart AI Document Chatbot with RAG

Python, FAISS, sentence-transformers, PyTorch, Gradio

RAG with FAISS retrieval, sentence embeddings, and an open-LLM path (e.g. Gemma-2B) documented in the public README; notebooks and a Hugging Face Space demo. Optional API keys for local experiments.

Emotion Recognition Project

Python, PyTorch, EfficientNet, OpenCV

Advanced emotion recognition system using EfficientNet-B0 with comprehensive model optimization pipeline. Features multi-phase development including model selection on FER-2013 dataset, fine-tuning on RAF-DB dataset, and dynamic INT8 quantization for mobile deployment.

AI Recipe Finder

Python, Whisper, Groq, Gradio

Voice-powered recipe app: say ingredients aloud and get AI-generated recipes. Uses Whisper for speech-to-text and Groq (or local FLAN-T5) for generation, built with Gradio and deployed on Hugging Face Spaces.

Tree Classifier

Python, Machine Learning, Vercel, Web, Mobile

Interactive web and mobile app for tree species classification using machine learning. Live demo on Vercel.

SKILLS

Machine Learning

- Predictive Modeling
- Time Series Analysis
- RAG Architecture
- Model Deployment
- Prompt Engineering

Data Science

- Data Mining & Analytics
- EDA & Visualization
- Feature Engineering
- Statistical Analysis
- Dashboard Creation

Leadership

- Team collaboration (15+ people)
- Cross-cultural Collaboration
- Mentoring & Training
- Project Management
- Stakeholder Communication

Business

- Strategic Planning
- Data Analysis
- Process Optimization
- Financial Management
- Business Development
- Operations Management

SELECTED PUBLICATIONS

Microsoft-based Performance Goal Management System for Strategy Implementation — Bachelor's thesis, Häme University of Applied Sciences (2026).
Catalogue: [HAMK Finna \(vanaicat.144542\)](#); PDF on the portfolio site.

PERSONAL INTERESTS

-  **Hiking** - Nature exploration, outdoor activities
-  **Fitness** - Health and wellness, physical activity
-  **Food Exploration** - Culinary experiences, cultural cuisine
-  **Continuous Learning** - Technology trends, professional development
-  **Environmental Protection** - Green initiatives, environmentally friendly practices